



FORGING THE INTELLIGENCE INFRASTRUCTURE

Cloud · Quantum · Defense

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Legion is a global technology company delivering **strategic technology consulting** across three converging frontiers: **Cloud Computing, Quantum Technologies**, and **Defense Systems**.

Inspired by the discipline of the Roman *Testudo* formation, we build cohesive elite teams where strength lies in the system — delivering functional, robust solutions designed for real-world use.

MISSION

Build intelligence infrastructure for the next era of computing

PHILOSOPHY

Disciplined execution, precise architecture, full cycle delivery

SCOPE

Global operations across Cloud, Quantum & Defense

Three Pillars

Cloud

Data & Infrastructure at Scale

- Data & AI
- Cloud Architecture
- Infrastructure
- Recruitment
- Machine Learning

Quantum

Next-Generation Computation

- Quantum Machine Learning
- Post-Quantum Cryptography
- Quantum Algorithms
- SIQIA
- Error Correction
- Quantum Simulation

Defense

Autonomous Systems & Security

- Unmanned Aerial Vehicles
- Unmanned Ground Vehicles
- AI Command & Control
- Computer Vision
- Cyber Defense

Strategic Roadmap

	Q2 2026	Q3-Q4 2026	2028	2030+
Cloud	Foundation Core data pipelines, cloud infrastructure setup, first client engagements	Scale ML platform launch, multi-cloud operations, staffing expansion	Maturity Enterprise-grade AI products, managed services, global client base	Dominance Full-stack cloud-AI platform, strategic partnerships, revenue engine
Quantum	Research Lab setup, algorithm research, Universidad Nacional partnership, SIQIA semillero launch	Prototype First QML models, PQC implementation, error correction experiments	Application Industry-specific quantum solutions, simulation products, PQC-as-a-Service	Quantum Advantage Demonstrated advantage in target domains, commercial products
Defense	Concept UAV/UGV prototyping, CV research, cyber defense framework	Development Autonomous navigation, C2 systems, offensive/defensive cyber tools	Testing Field trials, multi-agent operations, PQC-secured comms integration	Deployment Operational systems, defense contracts, full autonomous capability

01 Cloud

Data & Infrastructure at Scale



Cloud Division

Data & AI

Data engineering pipelines, ETL/ELT, data lakes, analytics platforms, and AI-driven data products for enterprise-scale operations.

Cloud Architecture

Multi-cloud strategy, infrastructure design, migration, and optimization across AWS, GCP, and Azure environments.

Infrastructure

IaC, Kubernetes orchestration, CI/CD pipelines, observability stacks, and platform engineering for production-grade systems.

Recruitment

Technical staffing and talent acquisition for cloud, data, and engineering roles across global markets.

Machine Learning

MLOps, model training and deployment, feature stores, model monitoring, and end-to-end ML platform development.

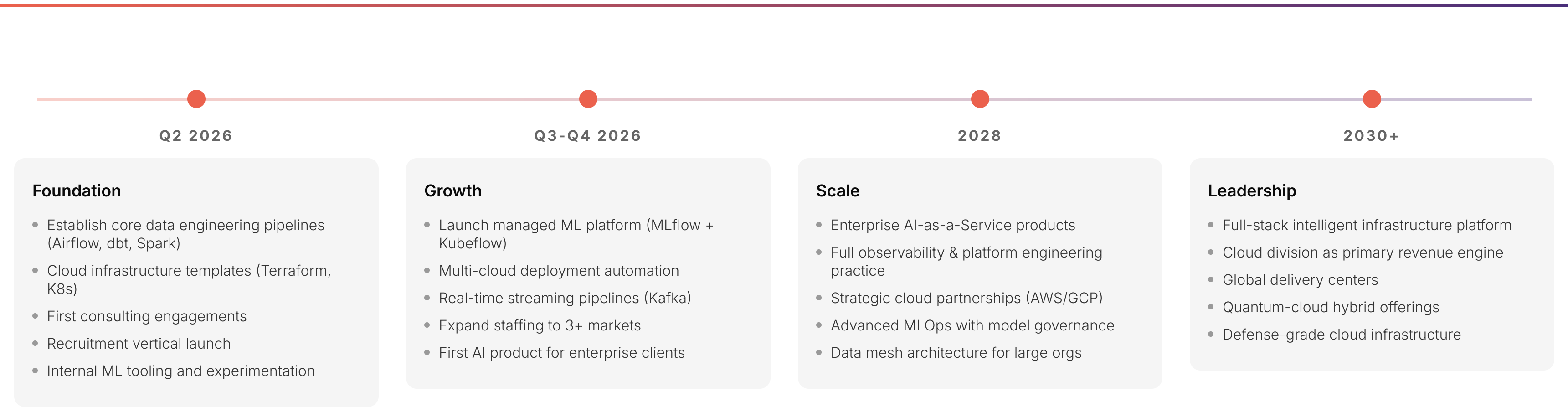
VISION

Become the backbone of intelligent infrastructure — enabling organizations to process, analyze, and act on data at any scale.

KEY TECHNOLOGIES

Apache Spark / Kafka / Airflow
Kubernetes / Terraform / ArgoCD
AWS / GCP / Azure
MLflow / Kubeflow / SageMaker
dbt / Snowflake / BigQuery

Cloud Roadmap



02 Quantum

Next-Generation Computation



Quantum Division

Quantum Machine Learning (QML)

Hybrid quantum-classical algorithms for optimization, classification, and generative modeling with quantum advantage.

Post-Quantum Cryptography (PQC)

Lattice-based, hash-based, and code-based cryptographic schemes resistant to quantum computational attacks.

Quantum Algorithms

Design and implementation of quantum algorithms for search, optimization, simulation, and linear algebra problems.

SIQIA

Research seedbed (semillero) in partnership with Universidad Nacional — quantum computing research, applied quantum information, and academic talent development.

Error Correction

Quantum error correction codes, fault-tolerant architectures, and noise mitigation for reliable quantum computation.

Quantum Simulation

Simulating quantum systems for materials science, chemistry, and fundamental physics research.

VISION

Pioneer practical quantum applications that solve problems intractable for classical computers — from cryptography to molecular simulation.

KEY TECHNOLOGIES

Qiskit / Cirq / PennyLane
NIST PQC Standards (CRYSTALS)
Variational Quantum Eigensolver
Surface Codes / Topological QEC
Quantum Chemistry (VQE/QAOA)

Quantum Deep Dive

Quantum Machine Learning

Hybrid quantum-classical computation

- Variational Quantum Circuits (VQC) for classification and regression
- Quantum Approximate Optimization (QAOA) for combinatorial problems
- Quantum Kernel Methods — leveraging high-dimensional Hilbert spaces
- Quantum Generative Adversarial Networks (qGAN) for data synthesis
- Barren plateau mitigation and trainability research
- Hybrid architectures: classical pre/post-processing + quantum core

Target domains: logistics optimization, financial modeling, drug discovery molecular screening

Post-Quantum Cryptography

Securing the post-quantum world

- CRYSTALS-Kyber — lattice-based key encapsulation (NIST standard)
- CRYSTALS-Dilithium — lattice-based digital signatures
- SPHINCS+ — hash-based stateless signature scheme
- Hybrid classical-PQC transition protocols
- PQC integration for TLS, VPN, and IoT endpoints
- Crypto-agility frameworks for seamless algorithm migration

Critical for defense communications, financial systems, and government infrastructure

Quantum Research Lines

SIQIA

Research Seedbed — Legion + Universidad Nacional

- Joint quantum computing research program with Universidad Nacional
- Applied quantum information and algorithm development
- Quantum sensing and precision measurement research
- Quantum communication protocols (QKD) for secure channels
- Academic talent pipeline — students to researchers to engineers

University-industry bridge: training the next generation of quantum scientists and engineers

Error Correction

Fault-tolerant quantum computing

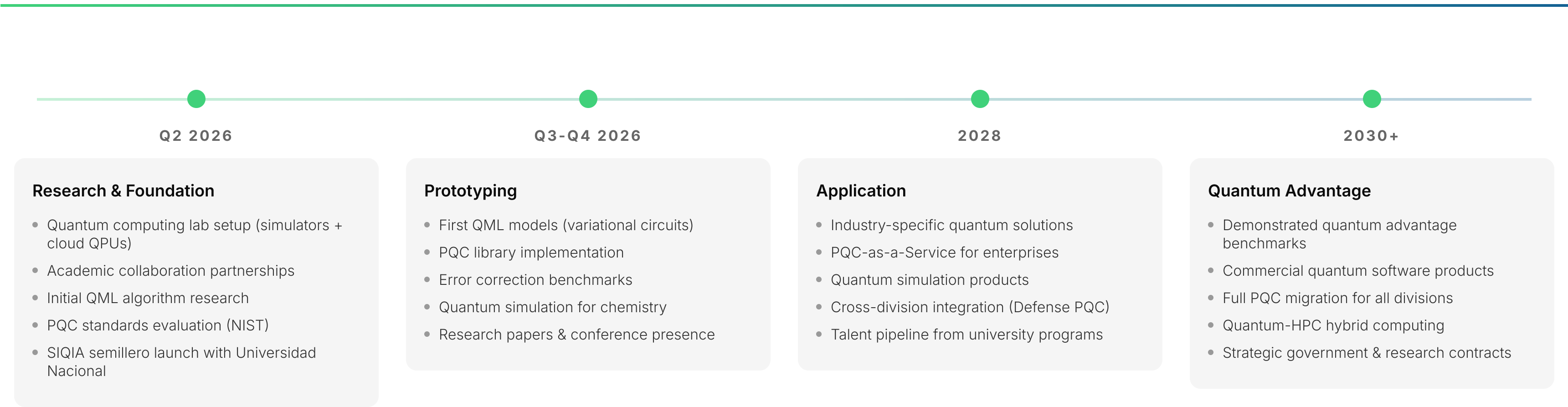
- Surface codes and topological protection
- Repetition and Steane codes
- Real-time syndrome decoding
- Noise characterization & mitigation (ZNE, PEC)

Quantum Simulation

Simulating nature at the quantum level

- Molecular energy surface mapping
- Materials discovery (superconductors, catalysts)
- Lattice gauge theories for particle physics
- Quantum dynamics of many-body systems

Quantum Roadmap



03 Defense

Autonomous Systems & Security



Defense Division

Unmanned Aerial Vehicles (UAV)

Autonomous drone platforms for reconnaissance, logistics, and tactical operations with AI-powered navigation.

Unmanned Ground Vehicles (UGV)

Autonomous ground platforms for patrol, cargo transport, mine clearance, and perimeter security operations.

AI Command & Control (C2)

Multi-agent coordination, decision support systems, and real-time tactical AI for autonomous robotics.

Computer Vision

Object detection, tracking, terrain mapping, satellite imagery analysis, and situational awareness via deep learning.

Cyber Defense

Offensive and defensive cybersecurity operations enhanced with PQC and AI-driven threat detection and response.

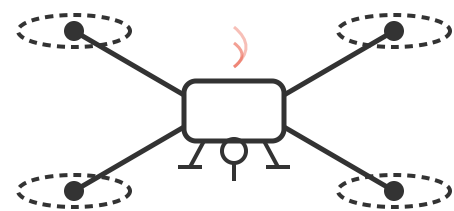
VISION

Build the autonomous defense systems of the future — where AI, robotics, and quantum-secure communications converge.

KEY TECHNOLOGIES

ROS2 / PX4 / ArduPilot
YOLO / SAM / Vision Transformers
Multi-Agent Reinforcement Learning
PQC-secured Communications
SIEM / EDR / AI Threat Hunting

Autonomous Platforms

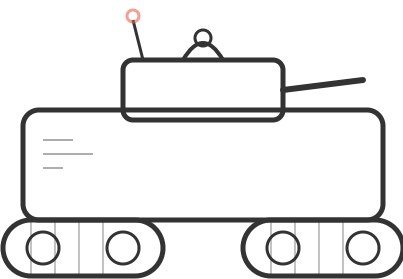


AERIAL PLATFORM

UAV Systems

Quadrotor and fixed-wing drones with autonomous navigation, real-time video feed, and AI-powered target detection for ISR missions.

PX4 / ArduPilot SLAM Navigation Real-time CV Swarm Capable GPS-denied Nav



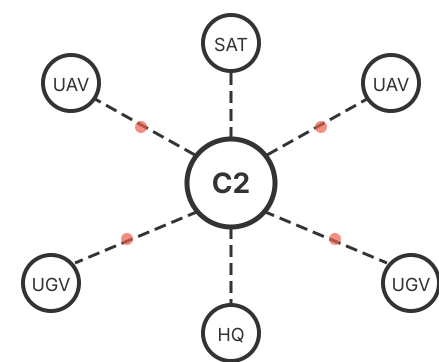
GROUND PLATFORM

UGV Systems

Tracked and wheeled autonomous ground vehicles for patrol, logistics, EOD, and perimeter security with multi-sensor fusion.

ROS2 LiDAR + IMU Terrain Mapping Remote Ops Modular Payload

Intelligence Systems

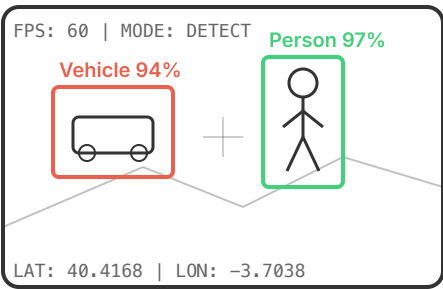


COMMAND ARCHITECTURE

AI Command & Control

Multi-agent orchestration platform that coordinates UAV/UGV swarms, fuses sensor data, and provides real-time tactical decision support.

- Multi-Agent RL
- Sensor Fusion
- Real-time Ops
- PQC Comms



PERCEPTION SYSTEM

Computer Vision

Real-time object detection, multi-object tracking, terrain classification, and satellite imagery analysis for full situational awareness.

- YOLOv8+
- SAM 2
- Vision Transformers
- Edge Inference

Cyber Defense

Offensive Operations RED TEAM

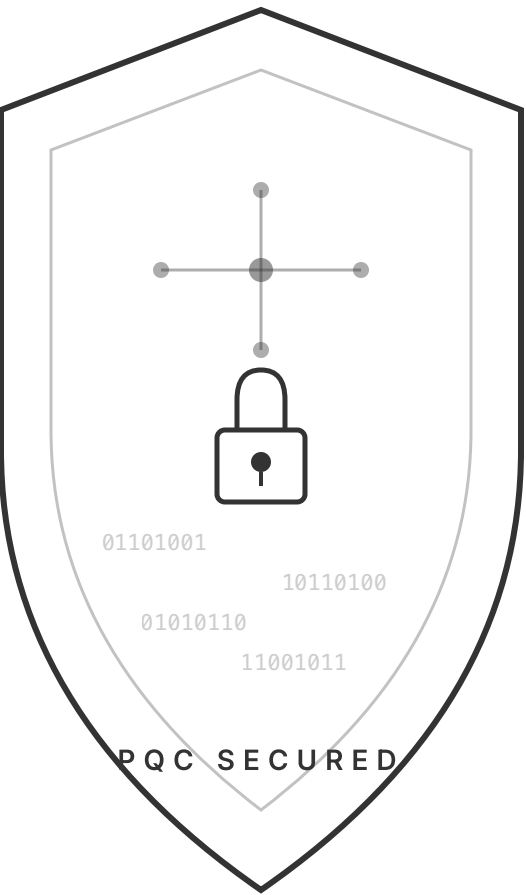
- Penetration testing and vulnerability assessment
- AI-driven attack surface discovery and exploitation
- Social engineering and phishing simulation campaigns
- Adversary emulation and red team exercises
- Zero-day research and responsible disclosure

Defensive Operations BLUE TEAM

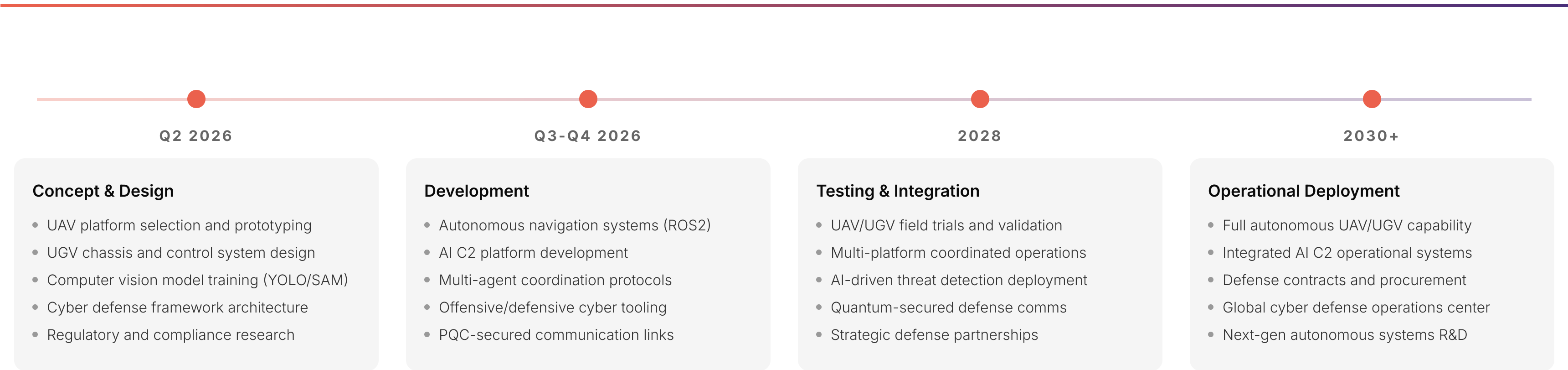
- AI-powered SIEM and anomaly detection (real-time)
- EDR/XDR deployment and threat hunting
- PQC-secured communication channels
- Incident response and digital forensics
- Zero-trust architecture implementation

Quantum-Enhanced Security

- Post-quantum cryptographic migration for all systems
- Quantum key distribution (QKD) for critical links
- Crypto-agility frameworks for algorithm rotation



Defense Roadmap





Let's Build the Future Together

FORGING THE INTELLIGENCE INFRASTRUCTURE

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